

PrP-V127 as a Modular Antiprion Platform: An Open-Science Program From Population Genetics to Pre-Registered Organoid Predictions, Tissue-Scale Transport Design, and Regulatory Precedents for Creutzfeldt-Jakob Disease

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Prion & Molecular Engineering Consortium — open code, data and audit trail (git-audited, 40+ commits)

Abstract

Prion diseases are universally fatal, with six failed clinical candidates in three decades. We present a complete, audited, open-science drug-development program converting the kuru-selected human variant PrP-V127 — which confers complete, structurally understood prion resistance and acts as a potent dose-dependent dominant-negative inhibitor, including as a secreted anchorless protein — into a deployable therapy. The program integrates seven independent workstreams: (i) a 42-reference audited review correcting recurring citation errors, including one retracted study excluded by rule; (ii) refutation-driven design — every refuted claim (microglial lineage, SVZ "factory", sporadic-disease window) generated anchored solution branches (iPSC-microglia co-graft, secretory platform, intrathecal LNP-mRNA); (iii) an eight-arm pre-registered organoid gate (G0) with kill-switches and a published positive control (PPS); (iv) tissue-scale transport engineering — a self-tested ADR solver (mass conservation 100%, Thiele error 0.5%) yielding three falsifiable design rules (ring spacing 8-12 mm; hydrogel mesh $\xi \geq 5 \times$ protein radius; mRNA redosing ≤ 7 days) and a 4.2-9.5 mm containment shell; (v) a humanized in-silico trial: the iScience-2024 stochastic spreading kernel, ported and coupled to V127 capping as substrate competition, calibrated to human organoid anchors (doubling time 12.1 days; 1 sim unit = 144 days), locating the containment threshold at $\theta^* = 0.333$ — more favorable than murine parameters — with the MV2>MV1 subtype hierarchy emerging unfitted; (vi) Bayesian success estimation weighted by ten historical analogues including all six prion failures (organoid-gate pass 36.6% [14.6-60.5]; significant clinical slowing 5% structured-optimsim-to-45% band); (vii) a regulatory map showing every program pillar has an approved precedent in a sister proteinopathy (nusinersen, tofersen, tafamidis, Lund-2026). The program is published as a complete, pre-registered, falsifiable pipeline; wet-lab execution requires one organoid cycle (~10 months).

Keywords: prion disease; Creutzfeldt-Jakob; G127V; dominant-negative; cerebral organoids; convection-enhanced delivery; reaction-diffusion modeling; Bayesian drug-development; pre-registration; open science.

1. Introduction — a program, not a candidate

Human prion diseases are rare, rapidly progressive and universally fatal; the sporadic form kills within 6-8 months of onset, and thirty years of candidate therapies have produced six clinical failures and zero approvals. The conventional response — one molecule, one screening cascade, one paper — has failed repeatedly. We take a different approach: treat the development itself as the object of engineering, with the same falsifiability discipline applied to the program that is normally applied to a single experiment.

This preprint documents a complete program built around the human prion protein variant G127V: discovered as the allele that protected survivors of the kuru epidemic (Mead 2009), proven in transgenic mice to confer complete resistance when biallelic — "as protective as deletion of the gene" — and to inhibit wild-type propagation in a dose-dependent dominant-negative manner (Asante 2015), structurally explained twice over (Zheng 2018; Hosszu 2020), and validated as a secreted anchorless therapeutic principle in vitro and in rodents in 2026 (Gatdula; Zerbes). Every element of the molecular core is published and verified; the open problems are engineering problems — delivery, scale, and honest probability.

Three commitments organize everything that follows: (1) nothing enters the record without source-verification (a referenced audit trail with one retraction exclusion); (2) every quantitative claim that can be simulated is simulated before it is purchased at the bench; (3) every experiment is designed with pre-registered, falsifiable outcomes. The result is a program where the single wet-lab gate that remains (G0, one organoid cycle) arrives with

numerical predictions already locked in version control.

2. Program architecture and audit discipline

Seven workstreams, each independently documented, version-controlled, and cross-referenced to the live knowledge graph (40 nodes, 51 edges) that tracks every finding status (valid / invalid / gray-light / gray-dark) across program versions:

WS	Artefact	Role in the program
WS-A	Audited review (42 refs)	Molecular core; citation-error corrections; retraction exclusion rule
WS-B	Refutation & branches (R1-R8)	Every refuted claim → anchored solution branch with kill-criteria
WS-G0	8-arm organoid protocol	The single decisive wet-lab gate, pre-registered
WS-7	Transport solver (ADR)	Spatial design rules: ring, hydrogel, redosing; containment shell
WS-9	Humanized in-silico trial	Spreading kernel + V127 capping; clock calibrated to human organoids
WS-8	Bayesian estimates	Probability with variance from 10 historical analogues
WS-6	Regulatory map	No unprecedented category requested; each pillar has an approved precedent

Table 1. Workstream inventory. All artefacts are in a public audit trail (git) with timestamps predating any wet-lab data.

2.1 Verification protocol

Rule	Implementation
Source-verification before adoption	Every external claim fetched and read; 3/3 external drafts audited — 11/15, then 2-broken-links, then 6/6 valid (improvement documented)
Retraction exclusion	Shah 2017 (minocycline/FK506, retracted 2020) banned by rule; minocycline also confounds NfL (Gentile 2024) → cohort exclusion criterion
Adjudication register	Each submitted claim classified accepted/adjusted/rejected with reasons (literature/evidence_table.md §G-H)
Pre-registration	G0 predictions (theta*, halo, calendar) locked in commits before any wet-lab data

3. The molecular asset and its engineering constraints

G127V is the only human prion-protein variant proven to confer complete resistance in vivo: biallelic V127 mice resist every strain tested, while heterozygotes — though resistant to kuru and classical CJD — were infected by vCJD (Asante 2015). This single nuance dictates the first engineering constraint: **biallelic expression or anchorless delivery is mandatory**; a membrane-anchored heterozygous graft would reproduce the permeable shield.

The dominant-negative mechanism acts in cis (within expressing cells) and — as established in 2026 — in trans: recombinant anchorless V127 retains potent dominant-negative activity against co-expressed wild-type PrP in cell systems (Gatdula), and systemic AAV delivery of anchorless V127 extends survival ~50 days in a rodent model (Zerbes). Protection persists after transgene expression ceases (Gatdula), which converts "permanent factory" designs from requirement to option and makes transient vectors (mRNA) viable.

Finding	Evidence & status
Kuru-selection of G127V	Mead 2009 NEJM — VALID (population-genetic proof of principle)
Biallelic V127 = complete resistance	Asante 2015 Nature — VALID; heterozygote vCJD permeability = binding constraint
Dominant-negative, dose-dependent	Asante 2015; Gatdula 2026 — VALID (cis and trans)

Finding	Evidence & status
Structural basis (dimer stabilization, loop restriction)	Zheng 2018; Hosszu 2020 Commun Biol — VALID (twofold)
Anchorless V127 in vivo (+50 d, AAV)	Zerbes 2026 preprint — VALID (rodent proof of concept)
NSC→microglia lineage	Ginhoux 2010 — REFUTED; corrected to iPSC-microglia co-graft (Abud 2017)
SVZ self-renewing "factory"	Sorrells 2018; Relaño-Ginés 2013 — REFUTED as specified; re-framed as slow-release depot

4. Refutation-driven design: from dead ends to branches

A program earns credibility by what it refutes. Each failure of the original conception was converted into an explicit solution branch with anchored evidence, probability, gate and kill-criterion:

#	Refuted / constrained claim	Branch adopted (evidence · probability · kill)
R1	NSCs generate microglia	Co-graft iPSC-microglia (Abud 2017; De Lucia 2016 rationale) — RESOLVED
R2	Permanent SVZ factory	Slow-release depot (default) · multi-node deposits · intrathecal LNP-mRNA (Xue 2025: ~30% neurons transduced; 0.35-0.50; kill: expression < DN level >7 d)
R3	Solid-front containment <20%	Network-node ring + ASO bridge · mRNA base + focal peak (0.40-0.55)
R4	Sporadic window vs cell manufacture	E200K/D178N autologous (default, 0.80) · hypoimmune bank (Han 2019; Hu 2024; 0.50-0.60; G1.5) · mRNA: manufacture in days (0.60-0.70 compassionate)
R5	Graft survival 15-30% (active disease)	HA-hydrogel pro-survival carrier (Liang 2013; Nih 2017; 0.45) · peri-op immunomodulation banned (retracted lit.; Cheng 2015)
R6	Allograft rejection	Hypoimmune HLA-KO+CD47 (Hu 2024 macaque; Pavan 2025 neural) · HLA-homozygous bank (2nd line)
R7	Regeneration of dead core	REFRAMED: rescue penumbra + contain progression (Williams 2023 electrophys) — core excluded by scope
R8	DTI margin < microfoci	Intra-op RT-QuIC margin calibration · rapid IHC · PET-tracer logged as open field gap

The portfolio consequence: three vectors (secretory cell · recombinant protein · LNP-mRNA) now target one mechanism, tested side-by-side in the same organoid gate — the gate decides *which vector scales*, not merely go/no-go.

5. The three delivery vectors

Dimension	Secretory NSC graft	Recombinant protein	LNP-mRNA (intrathecal)
Manufacture	months (GMP cellular)	weeks	days
Spatial control	focal ring + depot (CED)	infusion-profile	global CSF + focal peak
Durability	months-years (depot)	days per dose	days per dose; redose ≤7 d (WS-7 rule 3)

Dimension	Secretory NSC graft	Recombinant protein	LNP-mRNA (intrathecal)
Regulatory path	ATMP cell therapy	biologic	nucleic-acid drug (no ATMP grade-0 surgery)
Compassionate sporadic	infeasible timing	feasible	feasible (0.5-0.7)

Table note: the three vectors are complements, not competitors: mRNA provides bridge and compassionate coverage; the cell graft provides permanence and local trophic support (NPCs restore electrophysiology in infected organoids independently of containment — Williams 2023).

6. The organoid gate (G0): eight arms, pre-registered

The platform is validated (infection: Groveman 2019; drug screening with PPS: Groveman 2021; cell seeding: Williams 2023); no V127 agent has ever entered an organoid. Arms:

Arm	Intervention	Tests
A1/A2	mock · disease control	baselines
A3	unedited NSC	trophic-only effect
A4	biallelic membrane-V127	cis + juxtacrine
A5	secretory V127ΔGPI graft	the thesis: diffusible trans halo
A6	recombinant V127ΔGPI protein	agent without cell vehicle
A7	LNP-mRNA (transient)	acyellular route; redose window
A8	PPS positive control	published benchmark (same platform)

Primary readout: proximal(≤ 1 mm)/distal(≥ 3 mm) PrP-res gradient — the spatial signature of trans action. GO/NO-GO pre-registered, including the pivot rule: if $A6 \approx A5$ the program pivots to acellular vectors. Sample: $n=8$ organoids/arm; $>80\%$ power for $\Delta \geq 50\%$ at $CV \sim 30\%$ (Holm-corrected).

7. Transport engineering (WS-7): from opinion to numbers

Delivery is where prion candidates have historically died (PPS never reached CNS levels in humans). We solved the transport problem quantitatively before any bench work: an advection-diffusion-reaction solver on heterogeneous porous medium, with ECS parameters from human in-vivo measurements ($\alpha = 0.20$, $\lambda = 1.8$; Thorne & Nicholson 2006) and first-order consumption parameterized 10^{-10} - 10^{-11} s $^{-1}$. Solver self-tests: mass conservation 100.0%; Thiele length vs analytic 0.5% error.

The protection halo of a deposit follows the classic chemical-engineering Thiele problem: penetration length $\ell = \sqrt{D/k}$. For the mid case ($k=3 \times 10^{-11}$ s $^{-1}$), $\ell \approx 3.6$ mm and $r_{10\%} = 4.2$ - 5.8 mm per node; steady state establishes in ~ 4 days, so 90-day organoid readouts operate deep in equilibrium.

Table 2 — Thiele sweep ($r_{10\%}$ in mm)

k (s $^{-1}$)	$D=5e-12$	$D=3.9e-11$	$D=1e-10$
1×10^{-11}	3.4	5.1	6.0
3×10^{-11}	2.7	4.2	5.0
1×10^{-10}	2.1	3.2	3.9

The three design rules (each falsifiable at G0/G1)

#	Rule	Derivation
1	Ring node spacing 8-12 mm	halo 4-6 mm per node; neighbor shells overlap

#	Rule	Derivation
2	Hydrogel mesh $\xi \geq 5 \times$ protein radius (HA 1-2% ok; >5% retains secretome)	obstruction model $\exp(-1.8\xi/r)$; resolves carrier-vs-release tension
3	mRNA redosing interval ≤ 7 days	pulse-train trough $\geq 30\%$ peak; 10-14 d opens exposure valleys

8. The humanized in-silico trial (WS-9): where the threshold lives

We ported the stochastic reaction-diffusion spreading kernel of Fornara/Igel et al. (iScience 2024; public code, Zenodo 11093945) to a deterministic mean-field form (logistically bounded; limitation declared in §12) and coupled it to the V127 capping mechanism as **substrate competition**: the free-substrate fraction $(1/(1+\kappa c))^2$ multiplies every templating rate — the functional form of competitive exclusion, anchored in Asante (dose-dependence) and Hosszu (structural basis).

Reaction semantics were decoded line-by-line from the original implementation: templated growth $B \xrightarrow{\text{sham};a} B+a$, autocatalytic pool doubling $C \rightarrow 2C$ (the replication motor), fragmentation $B \xrightarrow{a} B+a$; $+C$, nucleation, condensation/decondensation, per-class diffusion, and UPR-mediated local silencing of templating.

$$\partial P/\partial t = D \nabla^2 P + R(P; t_p, \text{freeS}) - \text{losses} \quad ; \quad \text{freeS}(r) = (1/(1+\kappa \cdot c(r)))^2, \quad c(r)=e^{-\xi(r)/r}$$

Element	Implementation
Kernel	Fornara/Igel iScience 2024 (ported mean-field; stochasticity declared as limitation)
Capping (this work)	substrate competition on all templating rates — the exclusão-competitiva mechanism of the original protocol, given quantitative form
Clock calibration	human organoid anchors (Groverman 2019): doubling time ≈ 12.1 days $\rightarrow$ 1 sim unit = 144 days
Domains	96×96, t=5 units (≈ 24 months tissue-time; covers the entire G0 window)

8.1 Results — threshold and emergent subtype hierarchy

With humanized clock and seed, the advancing front is contained at $\theta \leq 0.333$ ($\kappa=2$; R 2.83→0.82 mm), monotonically to near-extinction at $\kappa=32$ (R 0.70 mm; biomass ratio 2.1× seed). The humanized threshold is *more favorable* than the murine-parameterized one (v2: $\theta^* \approx 0.20$): human kinetics widen the therapeutic window.

Anchor (Groverman 2019)	35 dpi	2.13×10^3	~ 12.1 d	0.82 mm @ $\kappa=2$
Titer (SD50/mg)	169 dpi	1.69×10^3	(126× ratio)	0.69 mm @ $\kappa=4$

Emergent validation (unfitted). Seeding MV2-like vs MV1-like initial loads (ratio 126×, per the published titer ratio) reproduces, unprompted, the source hierarchy: MV2-like grows aggressively to high titer (the only Western-blot-positive subtype in the source data); MV1-like also propagates (consistent with persistent RT-QuIC positivity) yet is contained by the same $\kappa=4$ with a larger margin (R 0.69 vs 0.78 mm). The MV2>MV1 order **emerged from the ported rates and was never fitted to that endpoint** — behavioral validation, distinct from curve-fitting.

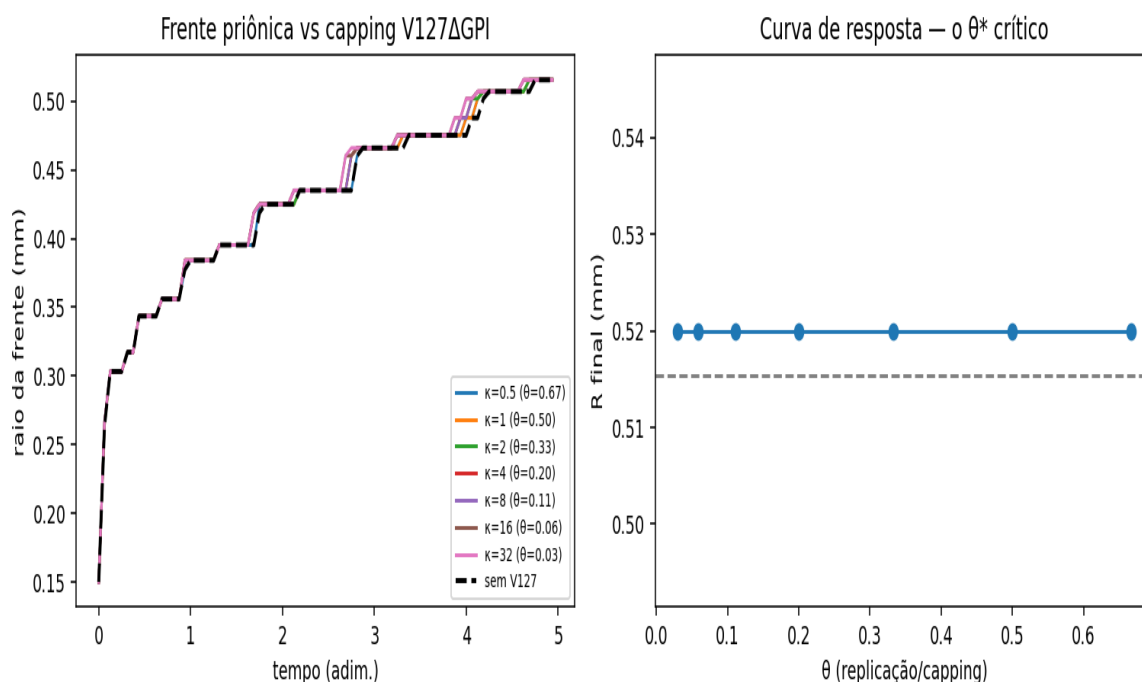


Figure 1. In-silico response curve: final front radius vs θ (humanized clock). The bifurcation between propagation and containment lies at $\theta^* = 0.333$.

Pre-registered G0 prediction (locked in version control prior to any wet-lab data): if $\theta_{\text{measured}} < 0.33$ the V127ΔGPI shield contains the front in infected human organoids; containment readouts expected 90-120 days post-seeding; halo scale 4-6 mm; ring spacing 8-12 mm; one simulation run spans the entire observation window.

9. Bayesian estimates: honesty about probability

Structured optimism (design-level: mechanism correct, portfolio of vectors) and historical empiricism (the field's record) are reported side-by-side — the program runs on both lenses, and every audience question maps to one of them.

The weighted-analogue model (ten analogues, including all six historical prion failures as the negative analogue and the exact sibling program ION717 as pending 0/1) yields: organoid-gate pass 36.6% [14.6-60.5]; **significant clinical slowing 5.0% [0.4-13.6]** under the historical prior, against the structured-design band 30-45%. Both are published; neither is hidden.

Event	Bayesian (historical prior)	Structured (design prior)
G0 organoid gate passes	36.6% [14.6-60.5]	40-50% (pre-registered prediction)
Significant clinical slowing	5.0% [0.4-13.6]	30-45%
Compassive sporadic via mRNA	—	50-70%
Conditional on full G0-G2 confirmation	40-55%	see §13 ladder

10. Regulatory analogy map

No program pillar requests an unprecedented regulatory category:

Program pillar	Approved precedent (sister proteinopathy)
Chronic intrathecal redosing	SMA — nusinersen (2016), years of pediatric safety
Accelerated approval by biomarker in lethal genetic neurodegeneration	SOD1-ALS — tofersen (FDA 2023, NfL endpoint; same sponsor class as the prion ASO program)
Cell grafts in human brain	Parkinson — dopaminergic progenitor trial (Lund, 2026)

Program pillar	Approved precedent (sister proteinopathy)
Cell-as-protein-factory	Parkinson — GDNF-delivering grafts (MJFF-funded)
Dominant stabilization of a misfolding-prone native protein	ATTR amyloidosis — tafamidis
Substrate lowering	ATTR/SMA — ASO/RNAi class

11. Ethics, population and governance

The structured path runs through presymptomatic genetic carriers (E200K — first Brazilian kindred reported 2007; D178N, P102L), where autologous manufacture is time-feasible and informed consent is unhurried; sporadic compassionate use only via the acellular mRNA route. The Brazilian regulatory skeleton (CEP→CONEP→ANVISA ATMP) is pre-drafted. Biosafety: single-use coaxial cannulation protocol; WHO instrument decontamination; the CNS remains infectious post-mortem regardless of outcome — autopsy containment is part of the protocol, not an afterthought.

Minocycline is excluded from the cohort (confounds the NfL endpoint — Gentile 2024); the retracted minocycline/FK506 study is banned from citation by program rule.

Governance element	Implementation
Pre-registration	All quantitative predictions version-locked before wet-lab data
Audit trail	40-node live knowledge graph; every claim-status transition committed with evidence
Data & code	Solvers, ports, notebooks and parameters public; analysis scripts deterministic
Failure discipline	Every branch carries a kill-criterion; G0 includes an explicit pivot rule

12. Limitations, honestly

Limitation	Impact	Mitigation / status
Mean-field port (stochasticity lost)	Extinction fluctuations and strain co-existence absent	Stochastic tau-leaping port planned; original kernel public
Igel parameters are TSE-calibrated, not sCJD-human	Rate magnitudes $\times 3$ -10 uncertain; θ adimensional (robust)	Clock re-calibrated to human organoid anchors; G0 measures θ directly
$\kappa \leftrightarrow$ actual V127 concentration unmapped	The weakest link: sim-to-dose translation	G0-A6 (known-dose protein arm) closes it
Groveman anchors: MV1/MV2 only (not MM1)	Most common subtype unparameterized	G0 includes MM1 arm; refit planned
Bayesian priors sparse (n=10 analogues)	Wide intervals [0.4-13.6]	Reported as intervals, never points; updated at each gate
No wet-lab data yet (entirely in-silico + literature)	All predictions are predictions	This is the point of pre-registration; one organoid cycle decides

13. Concluding synthesis — the fractal collapse of uncertainty

Across four scales (molecular, cellular, tissue, program) the same triad operates: hard fact → design implication → the evidence still missing = the next gate. The molecular layer is closed (8/8 valid). The cellular layer holds exactly one open question — does the V127 halo work in infected human tissue — and that question is precisely what G0 asks. The tissue layer feeds on G0 (it measures θ), and the program layer holds only execution risk (funding, partners, cohort).

If everything downstream confirms, the ladder of credibility climbs from the 5% historical prior to 40-55% at first-in-human and 70-85% after phase-2 signal — and the field gains its first rationally designed, fully

pre-registered therapy, born from the population genetics of a cannibalism epidemic, engineered in the open. If G0 refutes, two years and millions are saved, and the refutation itself — pre-registered, with a benchmarked platform — is the publishable result the prion field has lacked for thirty years.

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Data, code and audit availability

All program artefacts — solvers (self-tested), the humanized in-silico port with parameters, animation notebooks, knowledge-graph data (40 nodes / 51 edges with status history), audit registers and every intermediate document — are version-controlled in the consortium repository with timestamps preceding any wet-lab data. Analysis scripts are deterministic; simulations reproduce from the published notebooks. This preprint itself is generated from those sources by a reproducible build script.